TRANSFORMERS

References

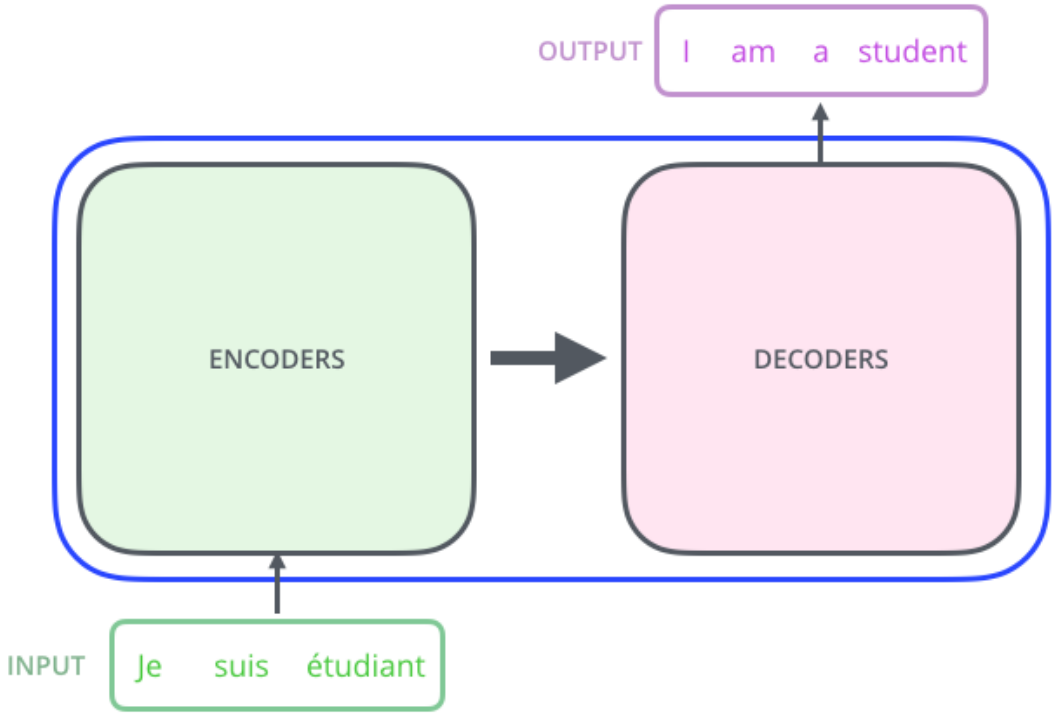
- A. Vaswani et al., "Attention is all you need", NeurIPS 2017.

- A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale", ICLR 2021.

- J. Alammar, "The Illustrated Transformer", https://jalammar.github.io/illustrated-transformer/
  - Source for most of the slides in this lecture

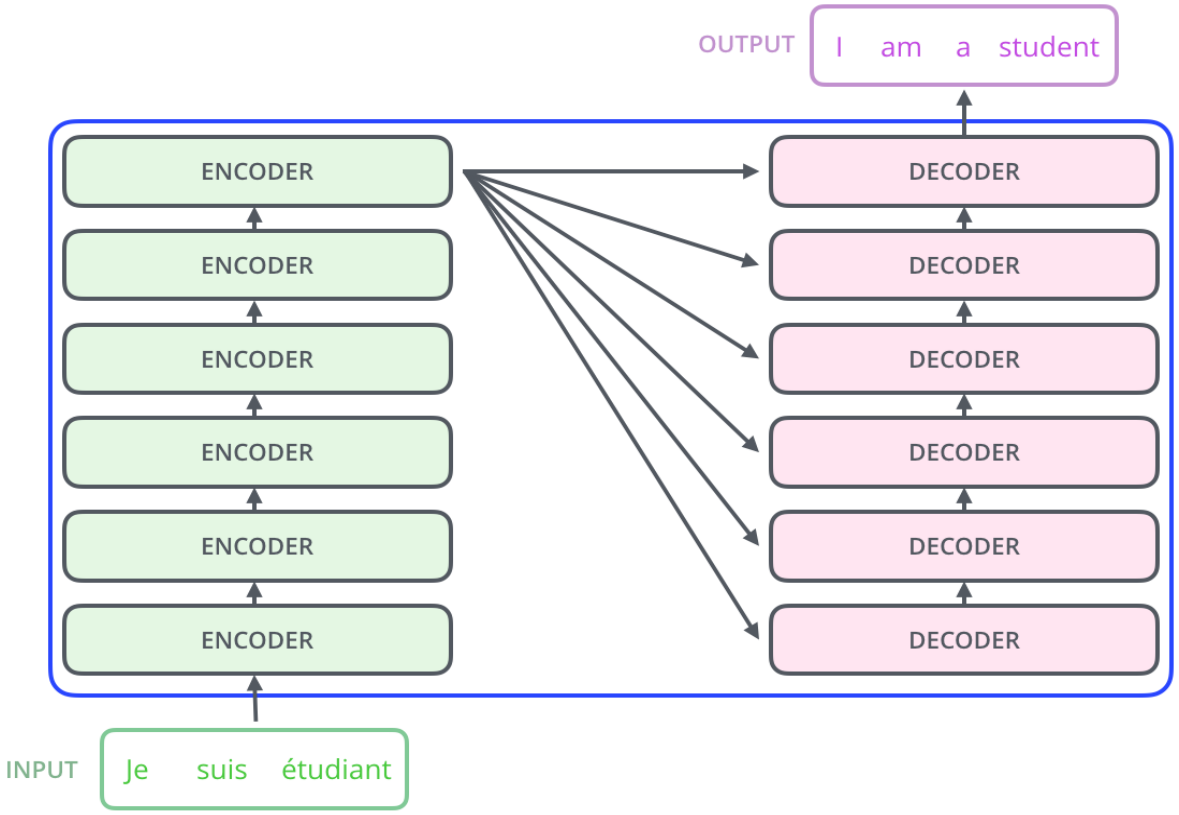
A High-Level Look

- An encoder-decoder architecture



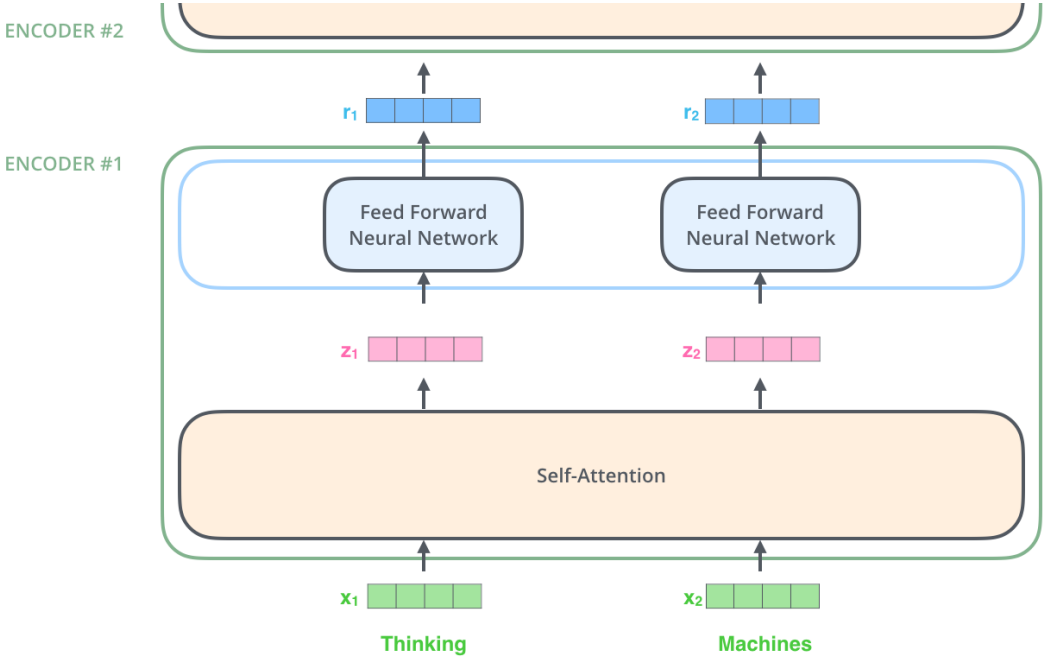
A High-Level Look

- Stacks of encoding/decoding blocks



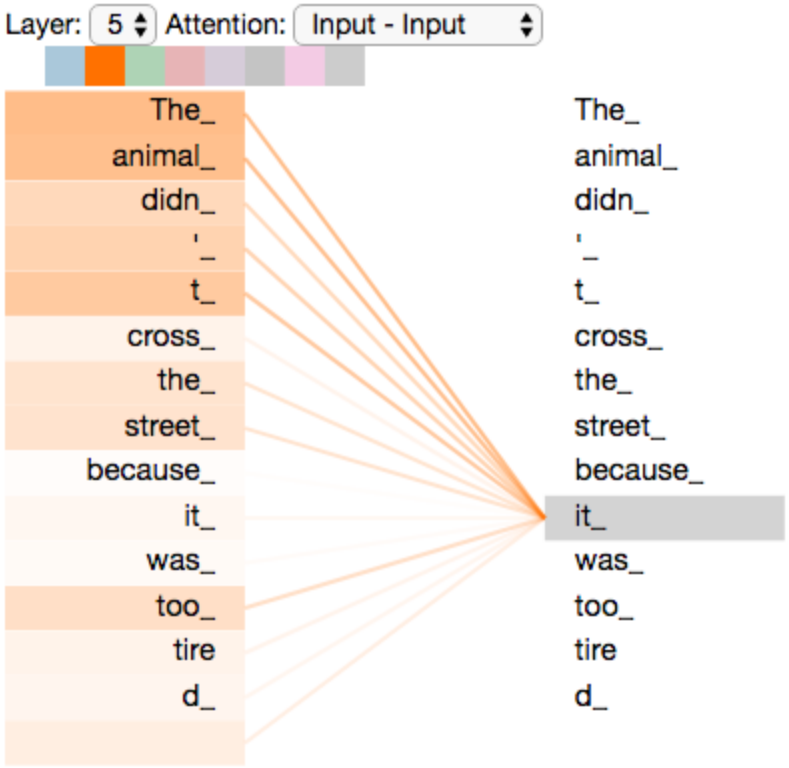
Encoder block

- A self-attention layer
  - Looks at other words in the input sentence as it encodes a specific word
- A feed-forward neural network
  - Applied independently to each word



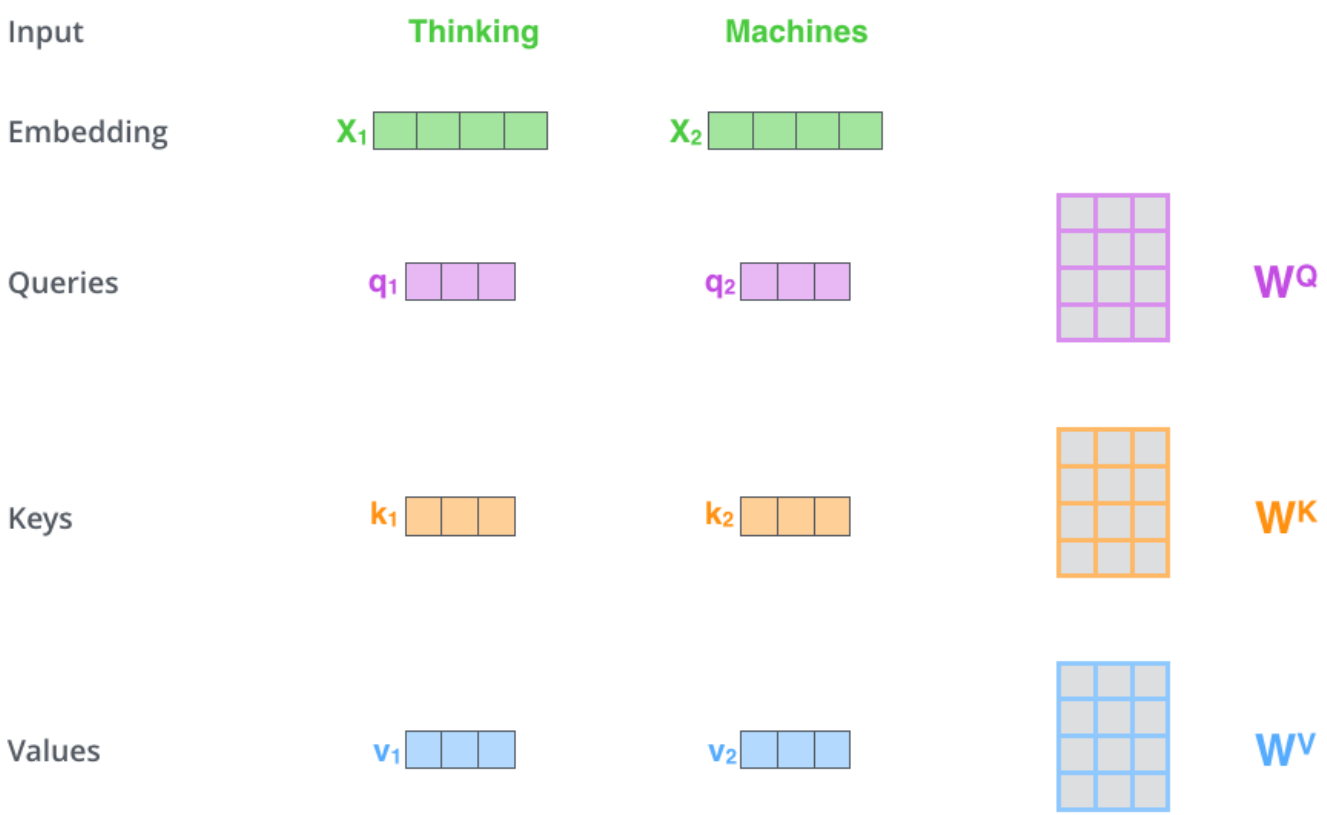
Self-Attention at a High Level

- Given the sentence:
  - *"The animal didn't cross the street because it was too tired"*
- What does *"it"* in this sentence refer to?
  - Is it referring to the *"street"* or to the *"animal"*?
- Self-attention allows the model to associate *"it"* with *"animal"*



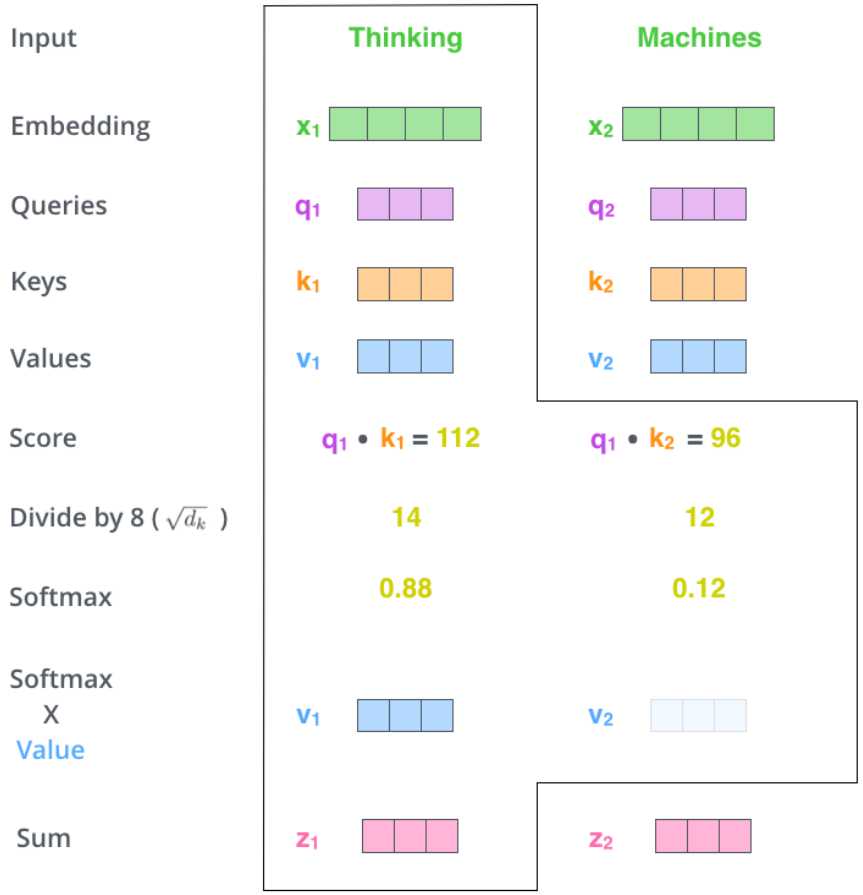
Self-Attention in Detail

- Create three vectors for each input vector: Query, Key, and Value

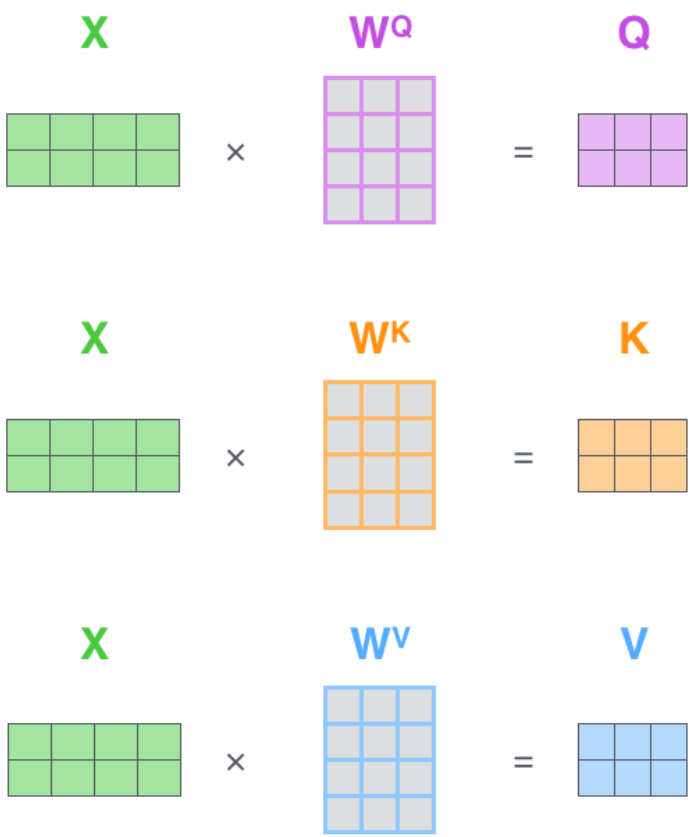


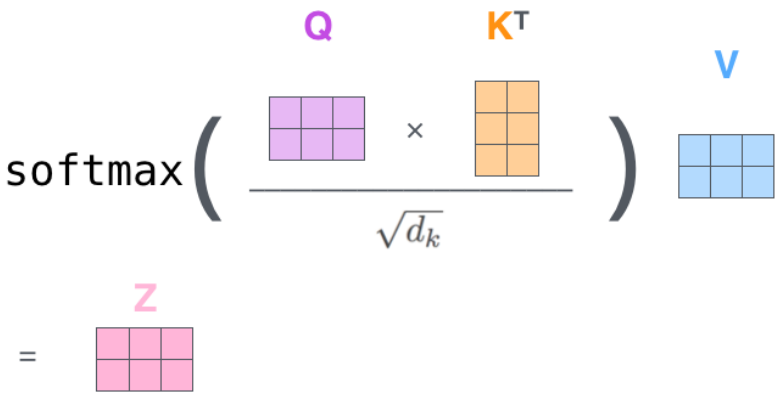
Self-Attention in Detail

- Query
  - What am I looking for?
- Key
  - What do I have to offer?
- Value
  - What will I pass forward?

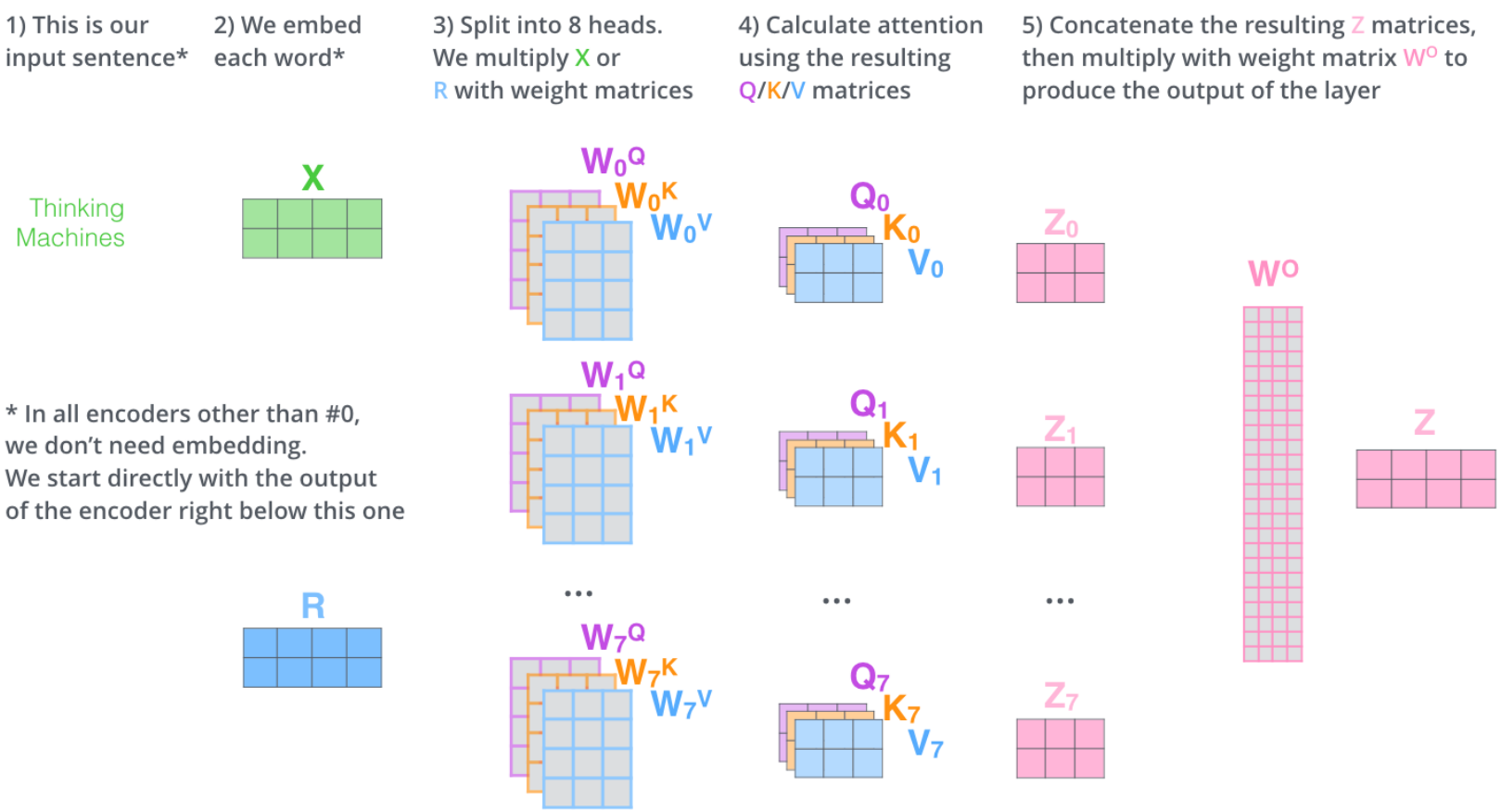


Efficient implementation

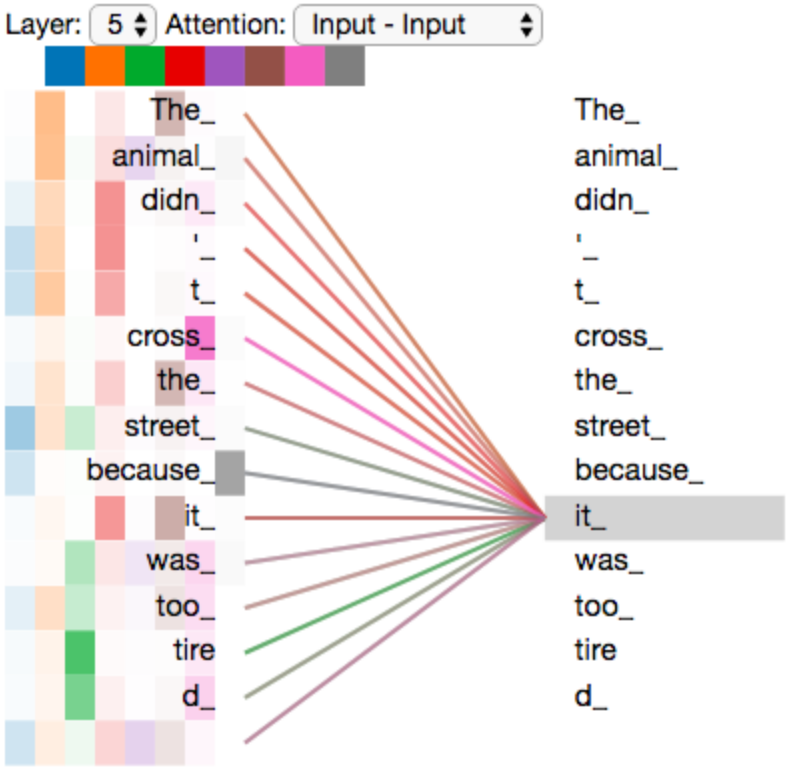


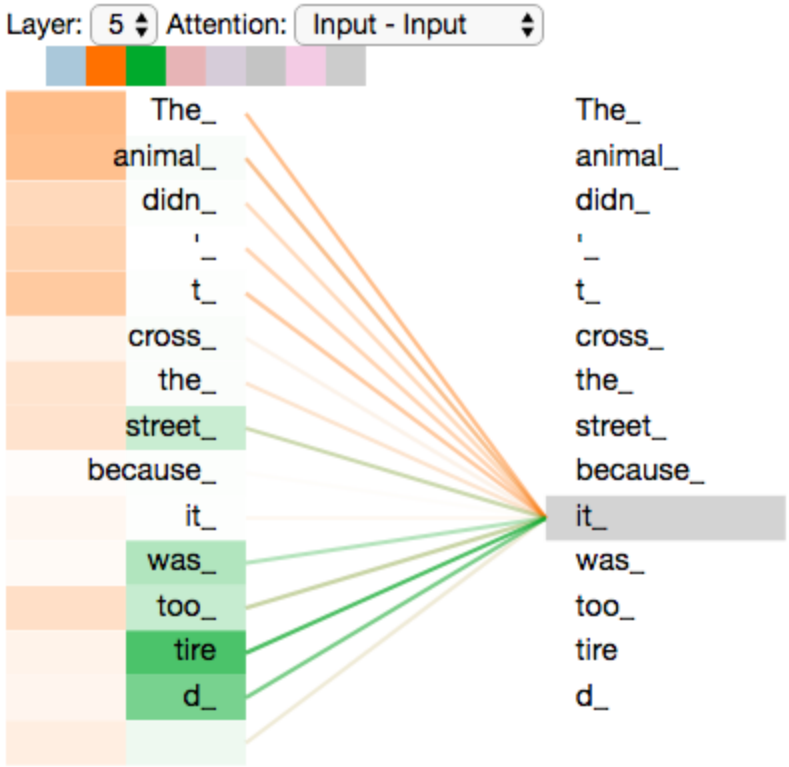


Multi-headed self-attention



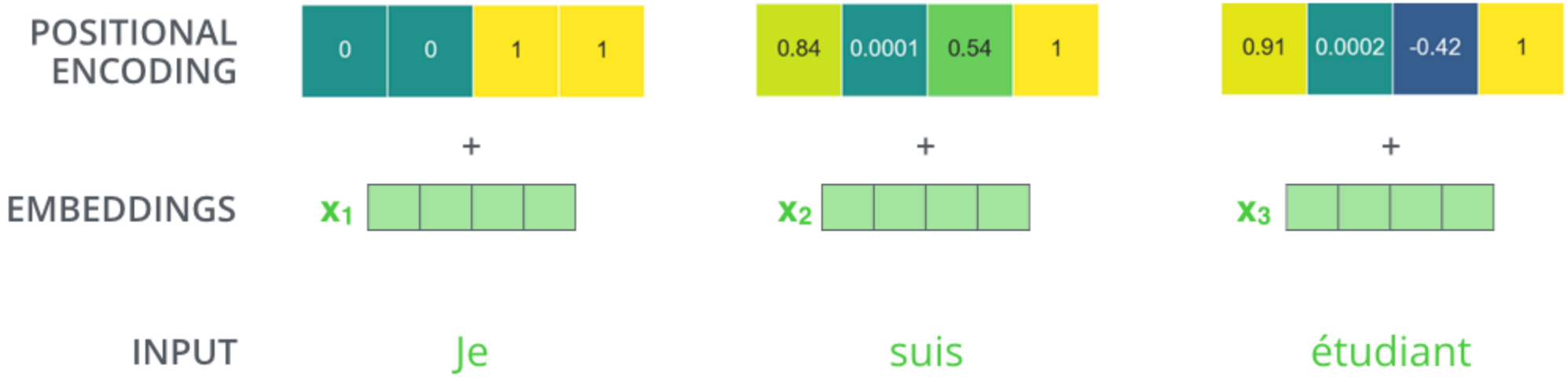
Multi-headed self-attention



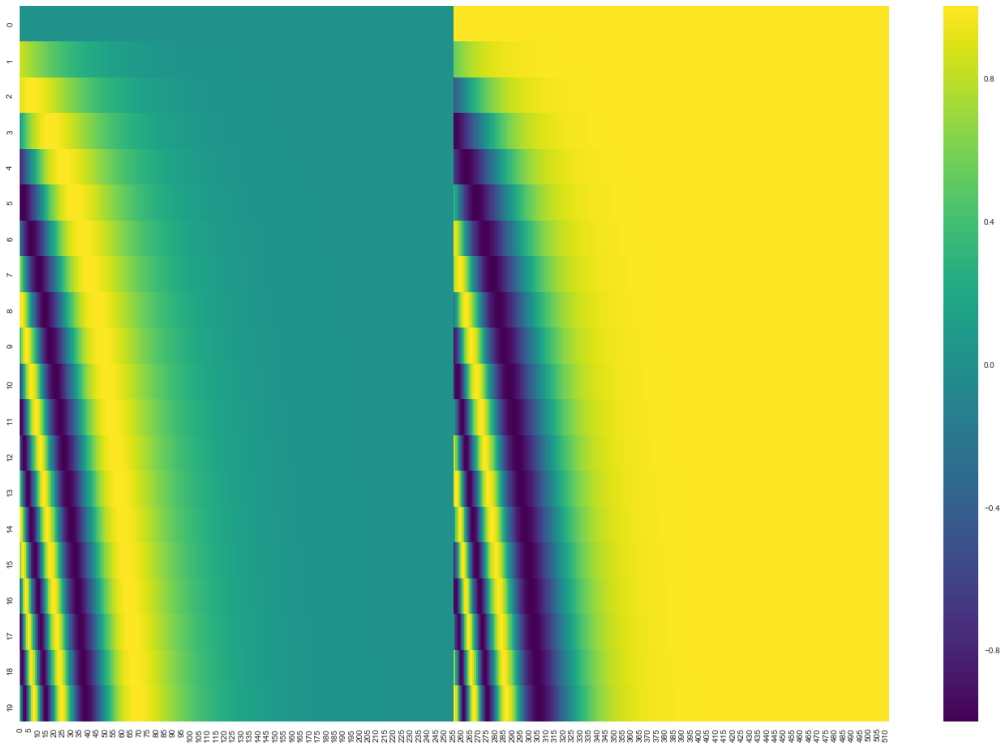


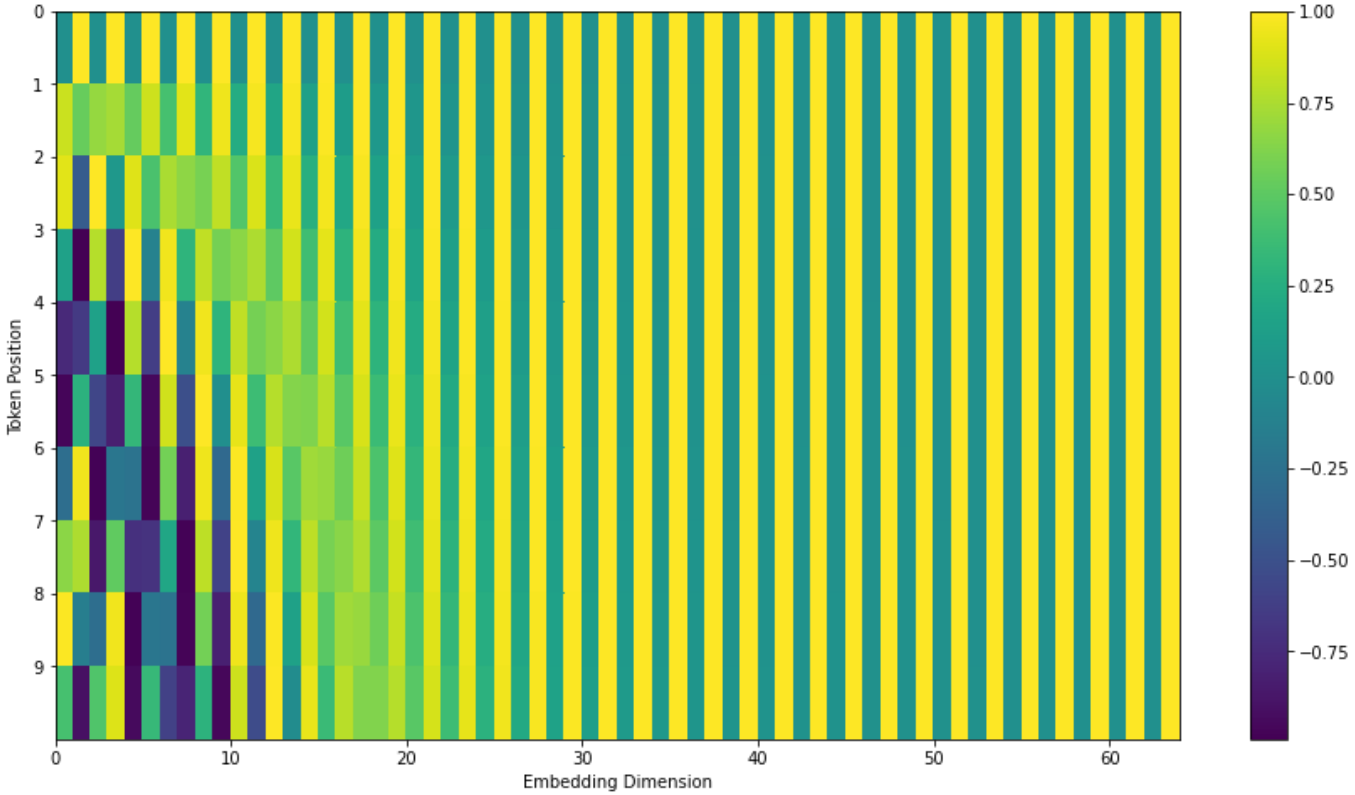
Positional Encoding

- Self-attention is permutation-invariant
- Solution: add vectors with a specific pattern to help the model determine the position of each word

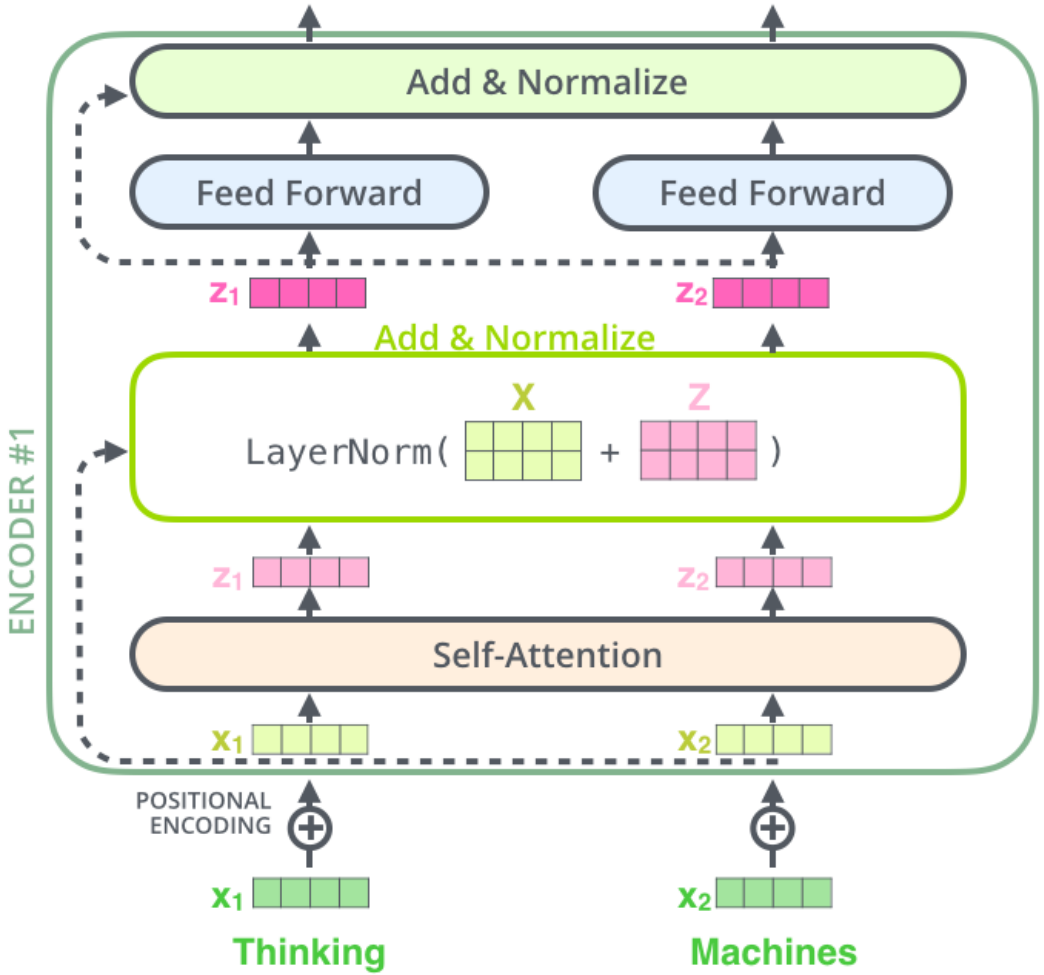


Positional Encoding

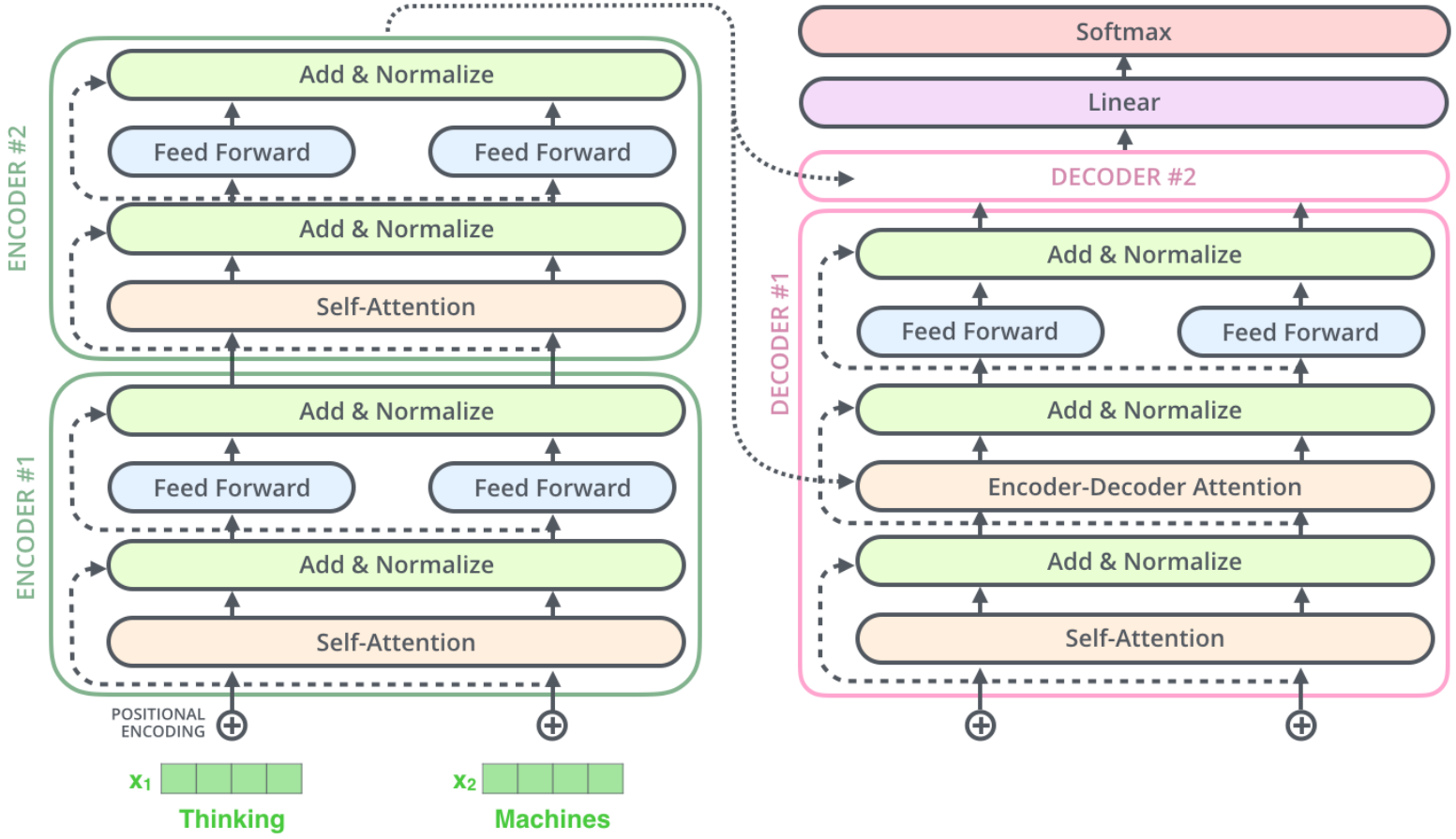




Detailed encoder block

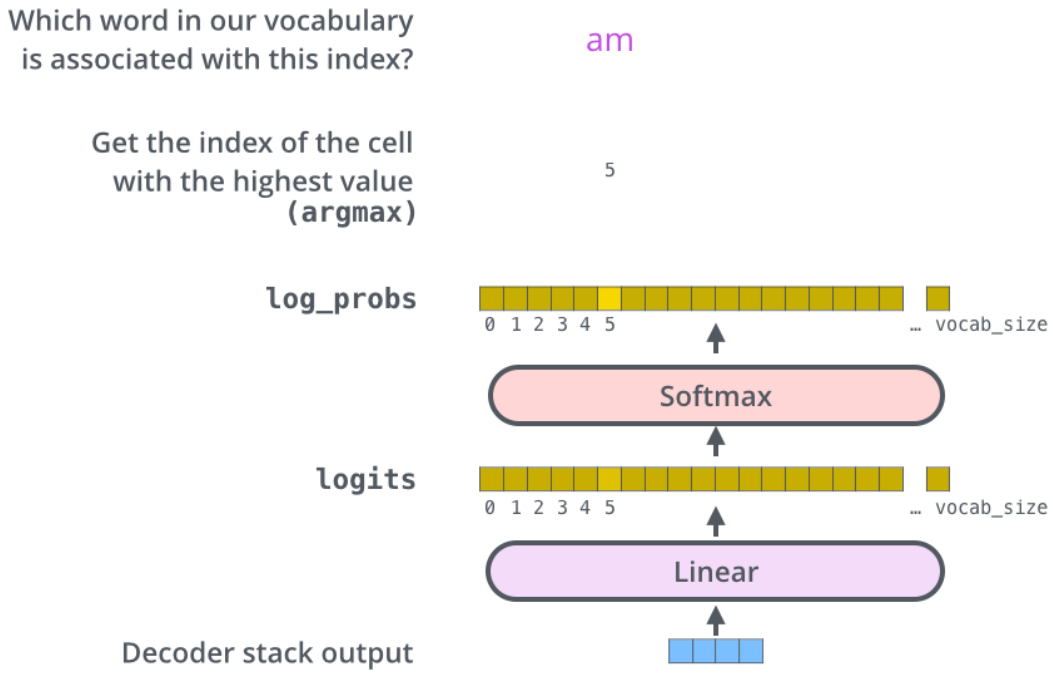


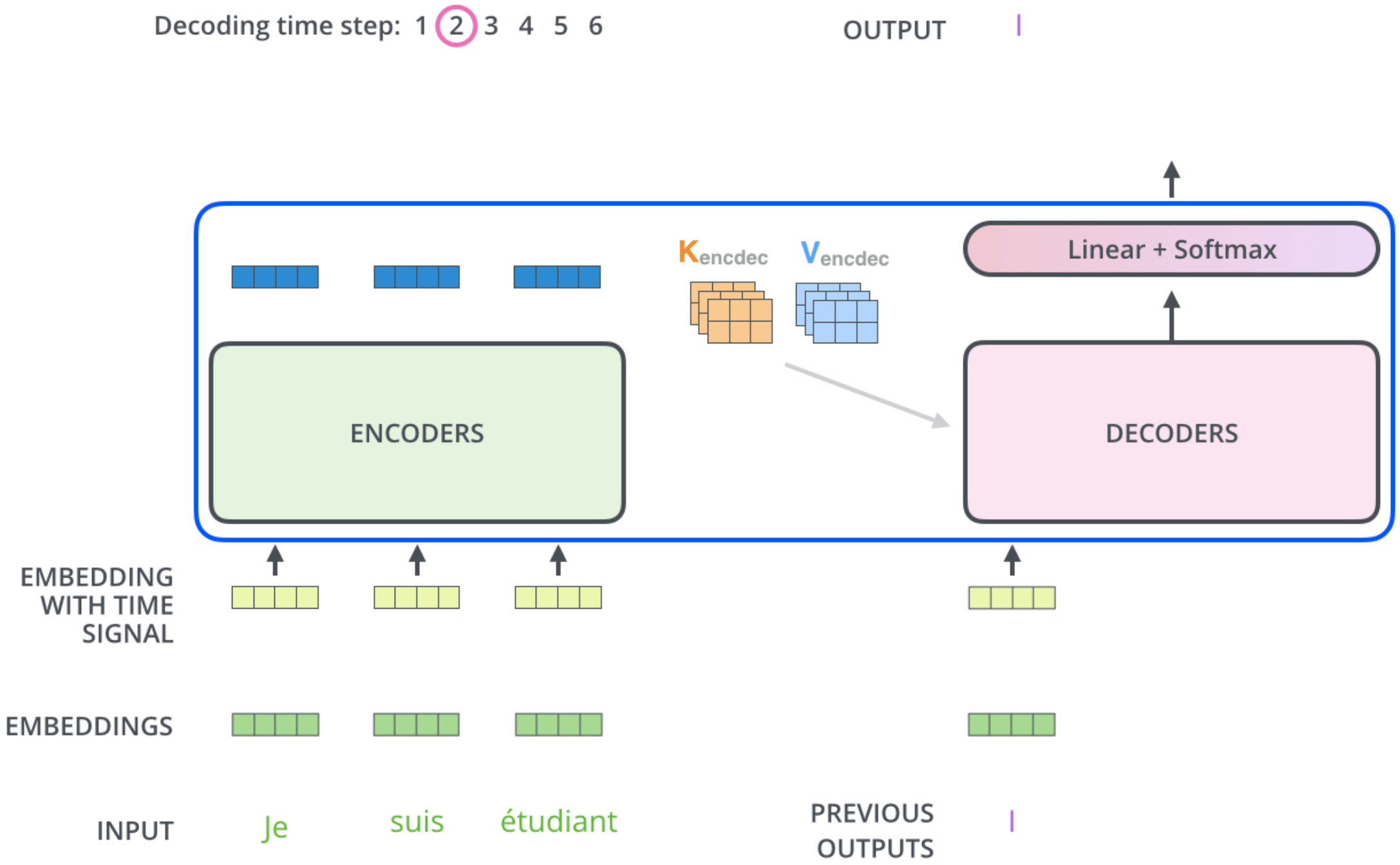
Decoder block



Decoder block

- Self-attention layer is only allowed to attend to earlier positions
- Cross-attention between input and output
  - Queries from the layer below it, Keys and Values from the encoder output
- Multi-class classification for word prediction




Decoding time step: 1 2 3 4 5 6
OUTPUT
I
Linear + Softmax
Kencdec
Vencdec
ENCODERS
DECODERS
EMBEDDING WITH TIME SIGNAL
EMBEDDINGS
INPUT
Je
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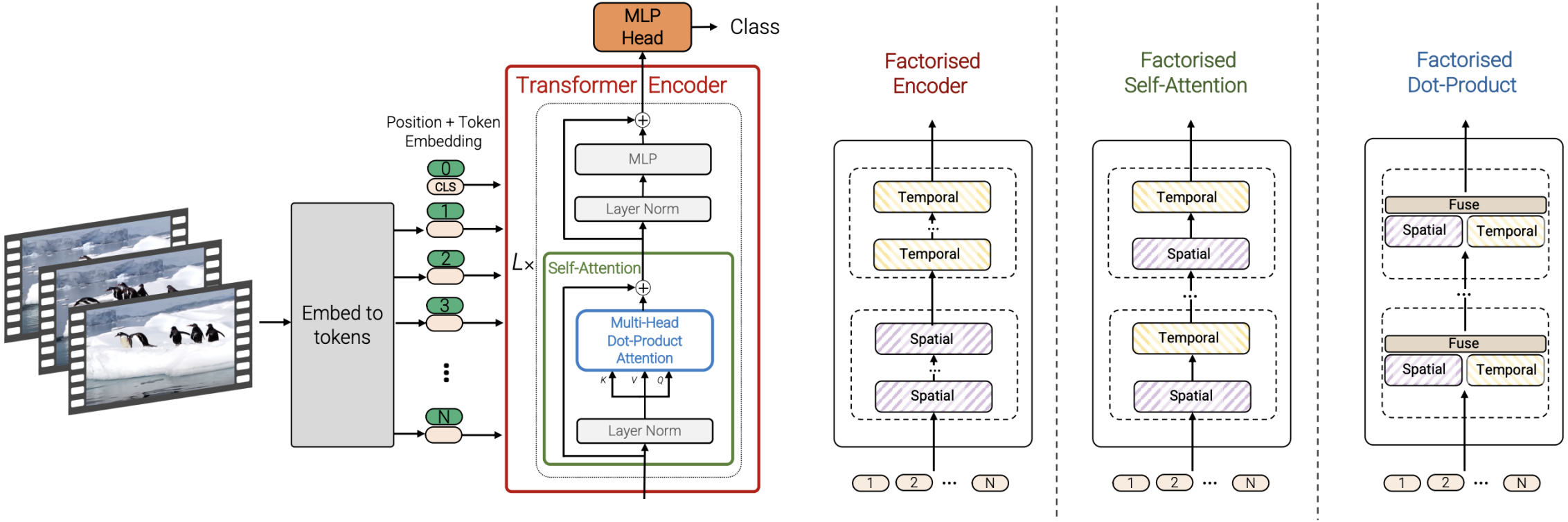
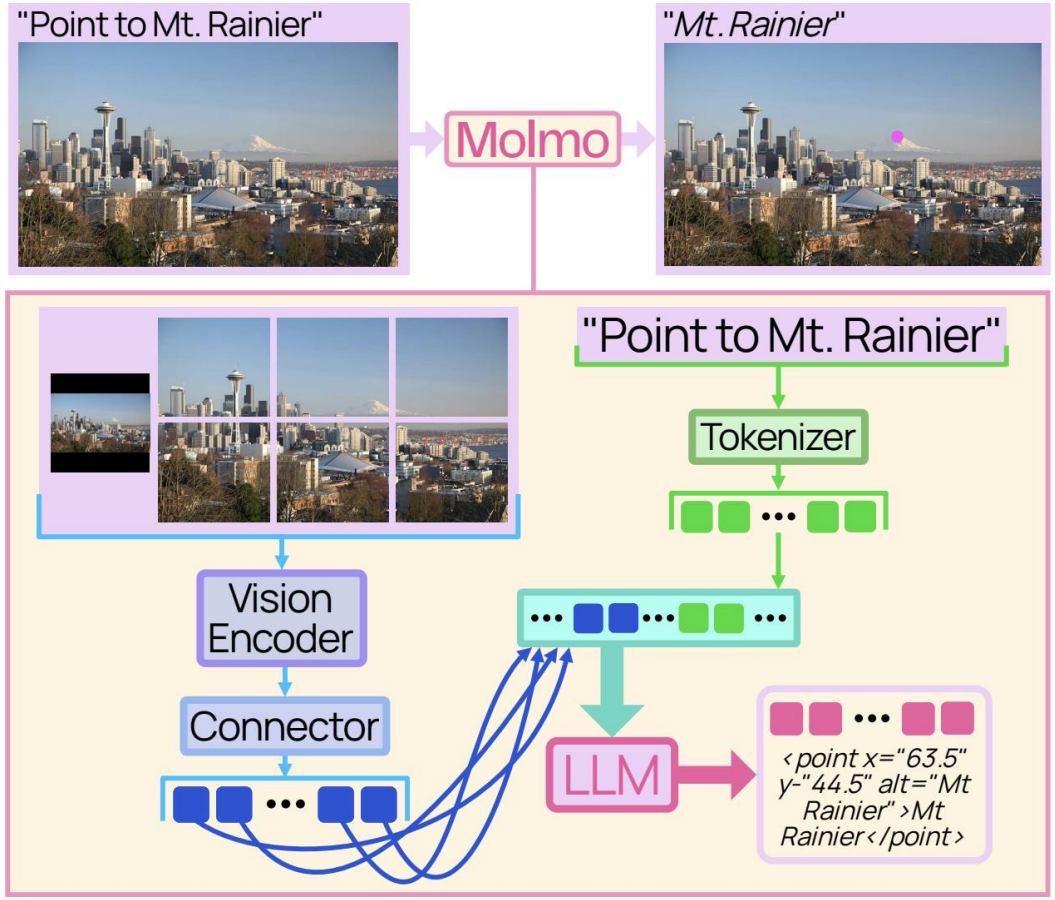
PREVIOUS OUTPUTS
I

Vision Transformer

Video analysis

- Arnab et al., "ViViT: A Video Vision Transformer", ICCV 2021.



Large Vision-Language Models

- M. Deitke et al., "Molmo and PixMo: Open Weights and Open Data for State-of-the-Art Vision-Language Models", CVPR 2025.